DAY 2 · 20 OCTOBER 2026

MIPT
MOSCOW INSTITUTE OF PHYSICS AND TECHNOLOGY

ML in medical technology

Ensembles on clinical tables, interpretability, a raw ECG processed live, and what breaks after deployment.

MIPT International School · Malaysian cohort · Aadi (Rituparn Shukla)

DAY 2

Today

01	Recap and HW1 debrief	10 min
02	Stronger models for biomedical tables	30 min
03	Interpretability	20 min
04	Biomedical signals: ECG code-along	45 min
05	Devices and healthcare systems	25 min
06	Choosing the operating point	15 min
07	Project release and defence format	20 min

HW1 debrief

The zeros. Glucose, blood pressure, skin thickness, insulin and BMI of 0 are impossible.

Who scaled before splitting? That is the single most common mistake in this course.

PPV answers: at 10% prevalence, 32.1%. At the Pima prevalence of 34.9%, 69.5%. Same test, twice the trustworthiness.

Median imputation inside the Pipeline should have *beaten* the raw-zeros pipeline. If yours did not, we will look at it during practice.

02

Stronger models for clinical tables

From one tree to many

Random forest: bagging

Hundreds of deep trees, each on a bootstrap sample of patients and a random subset of features at every split.

Averaging cancels the instability that made a single tree untrustworthy. Hard to overfit, easy to run, a sane default.

Gradient boosting : residuals

Small trees fitted in sequence, each one on what the previous ensemble got wrong.

Usually the strongest model on tabular clinical data. Needs more care: too many rounds and it memorises.

On tables of a few hundred to a few thousand patients, and they fit on your laptop.

Two lines, two strong models

```
from sklearn.ensemble import RandomForestClassifier, HistGradientBoostingClassifier

rf  = RandomForestClassifier(n_estimators=500, class_weight=
"balanced" , random_state=42)
hgb = HistGradientBoostingClassifier(max_iter=300, random_state=42)
# HistGradientBoosting accepts NaN directly — no imputer needed
```

Missing values are the normal state of clinical data. A model that handles them natively is a real advantage.

Still compare against HW1's logistic regression. **If the simple model is within one standard deviation, prefer it**

When one class is rare

01

Weight the classes

class_weight="balanced" makes a mistake on the rare class cost more during fitting. Free, and usually the first thing to try.

02

Stratify everything

The split and every CV fold. Otherwise a fold can contain almost none of the class you care about.

03

Read PR-AUC too

ROC-AUC stays optimistic when negatives dominate. Precision–recall AUC exposes how many alarms are false.

Resampling tricks come later. **Never resample outside the pipeline** , and never resample the validation fold.

Demo: does the fancy model actually win?

```
for name, m in {"logreg": logreg_pipe, "rf": rf, "hgb": hgb}.items():
    s = cross_val_score(m, Xtr, ytr, cv=cv, scoring="roc_auc")
    print(f"{name:7s} {s.mean():.3f} ± {s.std():.3f}")
# A gain of 0.01 with a standard deviation of 0.03 is not a gain.# Say so in your project instead of
declaring a winner.
```

A small grid search is fine and use **GridSearchCV inside the cross-validation**, never on the test set.

03

Interpretability

A clinician will not act on a number without a reason.

Who needs the explanation

The clinician has to decide whether to override the model. Without a reason, they either obey blindly or ignore it entirely.

The patient is entitled to know why they were flagged.

The regulator and your own team need to see that the model learned physiology rather than an artefact of how the data was collected.

The classic failure: a model that learned the **scanner** , the ward, or the ruler in the photograph and scored beautifully doing it.

Two honest tools

Coefficients

Only for linear models, and only meaningful on scaled features. Direction and size, per unit, holding the others fixed.

Permutation importance

Shuffle one column and re-score. The drop in performance is that column's contribution. Works for any model, measured on held-out data.

```
from sklearn.inspection import permutation_importance
r = permutation_importance(model, Xte, yte, n_repeats=30, random_state=42, scoring="roc_auc")
```

What importance does not mean

Correlated features share the credit. Radius, perimeter and area are nearly the same measurement and shuffling one leaves the others to carry the model, so all three look unimportant.

Importance is not causation. A feature can matter to the model because it is a marker of something else entirely or a consequence of the treatment.

Low importance is not "safe to drop" without re-fitting and re-validating.

Expect this question at the defence: **"Is your most important feature a cause of the disease?"** The correct answer is almost always no.

04

Biomedical signals: an ECG, live

Open ecg_codealong.ipynb. We write this together.

Five minutes of one heart

360

Samples per second. Nyquist: we can only see frequencies below 180 Hz.

108k

Samples in the recording — exactly five minutes of MIT-BIH record 208.

~500

Heartbeats in it, which we are about to find without a single label.

```
from scipy.datasets import electrocardiogram
ecg = electrocardiogram()
# 108,000 samples, in mV fs  = 360                          # Hz
```

One beat, three landmarks

P

Atrial depolarisation

A small bump before the spike. The atria contract and push blood into the ventricles.

QRS

Ventricular depolarisation

The tall, narrow spike the one the detector finds. Normally under 120 ms wide.

T

Repolarisation

The broad wave afterwards, as the ventricles reset for the next beat.

Everything we compute today is derived from **where the R peaks are** and **what the QRS looks like**.

Three kinds of noise, one filter

Baseline wander — breathing and electrode movement, below 0.5 Hz.

Mains interference — 50 Hz in Malaysia, 60 Hz in the Americas.

Muscle activity — broadband, high frequency, worst when the patient moves.

0.5 – 40 Hz

A third-order band-pass, applied with filtfilt so it introduces no phase shift — the peaks stay where they really are.

```
from scipy.signal import butter, filtfilt
b, a = butter(3, [0.5, 40], btype="band", fs=fs)
x = filtfilt(b, a, ecg)
```

Finding beats, then heart rate

```
from scipy.signal import find_peaks
import numpy as np

peaks, _ = find_peaks(x, distance=int(0.3 * fs), prominence=0.8)
rr = np.diff(peaks) / fs
# seconds between beats ok = (rr > 0.3) & (rr < 2.0)              # physiological limits hr    = 60 /
rr[ok].mean()
sdnn  = rr[ok].std() * 1000
d     = np.diff(rr)
rmssd = np.sqrt(np.mean(d[ok[1:] & ok[:-1]] ** 2)) * 1000
# 500 peaks · HR ≈ 103 bpm · SDNN ≈ 76 ms · RMSSD ≈ 104 ms
```

distance stops us counting the T wave as a second beat; prominence ignores small wobbles.

Where the simple detector fails

Three RR intervals of roughly **five seconds** appear in the record and around t = 45 s, 99 s and 214 s.

No heart paused for five seconds. Those are **artefact segments** : the signal is corrupted and the detector found nothing.

A production pipeline flags and excludes such segments. A naive one reports a heart rate of twelve.

Always plot the segment your metric came from. **Numbers computed from a signal nobody looked at are not evidence.**

Grouping beats with no labels at all

```
win = np.array([x[p - int(0.25*fs) : p + int(0.45*fs)] for p in peaks[1:-1]])
feat = np.column_stack([win, rr[:-1], rr[1:]])
# shape + RR before/after from sklearn.decomposition import PCA
from sklearn.cluster import KMeans
z = PCA(n_components=5).fit_transform(StandardScaler().fit_transform(feat))
lab = KMeans(n_clusters=2, n_init=10, random_state=42).fit_predict(z)
```

250 ms before each R peak to 450 ms after — one beat's worth of shape, plus the rhythm either side of it.

The small cluster is not noise

Roughly **370 beats in one cluster, 128 in the other**.

The small cluster's mean QRS is about **2.5× wider** at half amplitude, with a deeper negative deflection.

It arrives **early** and is followed by a **longer pause**.

Wide, early, compensatory pause: the textbook signature of a **premature ventricular contraction** .

104 ms

The RMSSD we computed earlier — inflated by exactly these ectopic beats. HRV must be measured on normal-to-normal intervals only.

Clustering found a clinical phenomenon without a single label. **Then a human named it** — and that step is not optional.

05

Devices and healthcare systems

Where the model lives, and what happens to it after you ship.

Three places a model lives

01

On the device

A wearable or a patch ECG. Milliseconds of latency, a coin cell of power, kilobytes of memory so small models, fixed point, feature-based.

02

At the bedside

Monitors and imaging consoles. More compute, but a hard real-time contract and an alarm budget: too many alerts and staff switch it off.

03

In hospital systems

Risk scores inside the electronic record: triage, deterioration, readmission. Plenty of compute, messy and delayed data.

The deployment target decides the model. **A wearable cannot run your random forest of 500 deep trees.**

The model degrades the day you ship it

A new hospital. Different population, different referral pattern, different prevalence — so a different PPV, with no code change.

A new sensor. Another manufacturer, another sampling rate, another electrode placement.

A new protocol. Clinical practice changed; the meaning of your features drifted with it.

Subgroup gaps. A model that performs well overall can fail for women, for the elderly, or for one ethnic group. **Report metrics by subgroup or you will not see it.**

Calibration: is a 0.8 really an 0.8?

80 of 100

Among patients given a score of 0.8, about 80 should truly have the disease. If only 40 do, the number is not a probability and it is a ranking.

Check it with a **reliability diagram** : predicted probability against observed frequency.

Summarise it with the **Brier score**.

Fix it with **CalibratedClassifierCV** on validation data, never on test.

AUC says nothing at all about calibration.

Software as a medical device

It is regulated as a device

In Malaysia the Medical Device Authority; in the United States the FDA; in Europe the MDR. Intended use, clinical evidence, risk management and post-market surveillance and all required.

A model that keeps learning after release changes the device. That needs a pre-agreed change plan.

Monitoring is part of the product

Track the input distribution, the positive rate, the alert volume and clinician override rate and not just accuracy, which needs labels you rarely get in time.

Decide in advance what triggers a retrain and who signs it off.

06

Choosing the operating point

Put a price on each mistake

$$\text{Cost}(t) = C_{FN} \cdot FN(t) + C_{FP} \cdot FP(t)$$

Sweep the threshold t, count the two error types at each value, and pick the t that minimises the total.

In cancer screening a missed tumour costs far more than a recall: $C_{FN} \gg C_{FP}$, so the threshold drops well below 0.5.

You do not need real currency. A ratio a clinician agrees with let's say **10 to 1** is enough, and it makes the assumption explicit.

Choose t on **validation data**. Report it on test, once.

Demo: the cost curve on Pima

```
C_FN, C_FP = 10, 1
ts = np.linspace(0.05, 0.95, 91)
cost = [C_FN*fn_at(t) + C_FP*fp_at(t) for t in ts]
t_star = ts[int(np.argmin(cost))]
# nowhere near 0.5
```

Every team must do this for its own project. **"We used 0.5" is not an operating point** — it is a default nobody chose.

07

Your project

Build and defend a clinical ML model. Full brief in the project deck.

Six tracks, two teams each

TRACK	DATASET	CLINICAL QUESTION	CHALLENGE	LEVEL
T1	heart-statlog	Does this patient have heart disease?	Only 270 patients — large CV variance	Easy
T2	ilpd	Liver disease from routine bloods	Imbalance, missing A/G ratio	Medium
T4	load_diabetes	How far will diabetes progress in a year?	Regression in clinical units	Medium
T5	SciPy ECG	Detect and characterise abnormal beats	No labels at all	Hard
T6	load_breast_cancer	Best diagnosis with at most 5 features	Feature selection and cost	Medium

Seven things every project must contain

The work

1. Problem framing: who uses it, what decision it supports, what an error costs. **2.** Cleaning done inside a Pipeline. **3.** A trivial baseline plus two real models, stratified 5-fold CV, mean ± std. **4.** A primary metric, chosen and justified — not accuracy by default.

The argument

5. An operating point: threshold from validation, confusion matrix on test. **6.** Interpretation, plus at least three misclassified patients examined by hand. **7.** Limitations: population, sample size, shift, what would be needed before clinical use.

T4 replaces 4 and 5 with error in clinical units and where the model is worst. T5 replaces 3 to 5 with features, clustering, and evidence that a cluster is real.

The next two hours

MINUTES	YOUR TEAM	ME
0–10	Pick a track, load the data	Approving tracks
10–40	EDA, cleaning inside the pipeline, dummy baseline	Round 1 — checking for leakage
40–75	Two real models, stratified 5-fold CV	Circulating
75–100	Operating point, confusion matrix on test	Round 2 — checking metric choice
100–120	Slide skeleton from the template	Exit ticket

Exit ticket: before you leave, tell me your primary metric and why, in one sentence.

How the defence runs

6

Minutes of talk. Six slides. I stop you at 6:30.

4

Minutes of questions, from me and from the room.

09:00

Submission deadline on the morning of the 21st, so every deck is pre-loaded on one laptop.

Run order is announced at the end of today. **Rehearse once, out loud, with a timer** because six minutes is shorter than you think.

Some sample questions

Why this metric, and not accuracy?
What happens to your PPV if prevalence drops to 5%?
Where in your pipeline could leakage have entered?
What does one false negative cost in your problem? One false positive?
Your two models differ by 0.01 AUC with a standard deviation of 0.03. Which do you pick, and why?
Is your most important feature a cause of the disease?

END OF DAY 2

Tonight: finish the argument

Interpretation, error analysis, limitations, six slides, one timed rehearsal. Submit by 09:00 tomorrow.

Defence · 21 October · 6 min talk + 4 min questions